

# A Study on Scalability and Performance of SQL and NoSQL Databases

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**Abstract**—Today's computing environments require a myriad of data storage systems to accommodate workloads such as real-time transaction processing and social graph data analysis, as well as event streaming and large-scale analytical queries. This heterogeneity has resulted in a myriad of database management systems. The primary difference between relational SQL databases and non-relational NoSQL databases involves engineering trade-offs of which there are a number of, and cannot be resolved in a uniform manner. This paper presents a thorough review of SQL and NoSQL databases along five dimensions: data modeling, schema, and query design; ACID vs. BASE; horizontal scalability and transaction throughput vs. latency based on the results of standardized tests. Five databases were selected as representatives of their respective styles: relational PostgreSQL, document-oriented MongoDB, key-value pair Redis, wide column Apache Cassandra, and graph Neo4j. Based on SQL Benchmark (YCSB) and a join-intensive OLAP workload, key-value stores are able to achieve a read throughput of up to 22× more than relational stores on the same hardware, while relational stores excel all NoSQL stores on complex five-table joins by 8× or more. To assist database designers and researchers, a practical decision-making model is provided.

*Index Terms*—relational databases, NoSQL, SQL, database benchmarking, ACID, BASE consistency, scalability, CAP theorem, workload analysis, data modeling.

## I. INTRODUCTION

The relational database model proposed by Codd in 1970 and later adopted by commercial systems such as Oracle, IBM DB2, PostgreSQL and MySQL became the most prominent data management systems for businesses. The principles of this model, declarative SQL, the ACID transaction model and normalization theory, provided a strong foundation for the development of reliable, verifiable and interoperable data infrastructures. For almost all workloads before the mid-2000s, relational systems were a good fit. As a result of many years of research and development, strategies for the optimization of indexing and concurrency management of requests have been developed that significantly reduce the number of disk input/output operations required for relational systems [1].

When internet-scale platforms emerged, data infrastructure requirements changed drastically. Services like Amazon's e-commerce site, Google's search engine, and Facebook's social network faced data challenges that traditional relational systems were never built to address: concurrent access from tens of millions of users, petabyte-scale datasets, globally distributed deployments requiring sub-second read performance, and data structures too irregular or deeply nested to be normalized into flat relational schemas without significant join overhead [2]. These demands gave rise to a new generation of storage engines, known as NoSQL ("not only SQL"), each of which addressing a specific subset of these challenges by

sacrificing one or more guarantees of the relational model [3].

Four principal NoSQL categories emerged: document stores (e.g., MongoDB, CouchDB), which store data as nested JSON structures; key-value stores (e.g., Redis, DynamoDB), which provide O(1) access to data via string keys; wide-column stores (e.g., Cassandra, Bigtable), which organize data into sparse partition-keyed tables optimized for time-series and analytical write patterns; and graph databases (e.g., Neo4j, Neptune), which treat entities and relationships as native objects queryable through path-oriented languages. In each category some aspect of relational capability, generally multi-entity join support or strict ACID guarantees, is sacrificed for horizontal scalability, flexible schema design or improved performance on certain access patterns [4].

Today's database architects work in an environment where the best storage solution depends fundamentally on the specific workload of the application and the consequences of that choice are significant. Deploying a relational database for a catalog service designed to span dozens of nodes introduces unnecessary complexity that a document store would naturally avoid. Alternatively, using a key-value store for your financial ledger means you lose the transactional integrity required for such applications. The current comparative research is still scattered. Most existing studies only consider one or two single engines under a single workload type, and the comparison across studies is

not reliable due to different hardware configuration, dataset volume, and consistency settings [5].

### A. Problem Statement

If we consider the database selection situation from four important dimensions, we observe a consistent gap in the current literature. There is no single published work that addresses all of the following simultaneously: (i) a well-defined taxonomy of all five major database paradigms; (ii) a reproducible empirical benchmark of representative engines under controlled hardware and dataset conditions; (iii) an evaluation of consistency models and their real-world operational consequences; and (iv) a prescriptive decision framework that maps workload characteristics to recommended engine families with quantitative support.

TABLE I. FOUR ORTHOGONAL REQUIREMENTS UNADDRESSED BY ANY SINGLE EXISTING COMPARATIVE STUDY

| Requirement                   | Rationale for Inclusion                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Structured Schema Enforcement | Relational databases enforce data integrity through schema definitions and constraints, reducing anomalies in transactional systems.                                          |
| Horizontal Scalability        | Modern web-scale applications require distributing data across commodity nodes; NoSQL databases are architected for this workload from the ground up.                         |
| Flexible Data Models          | Semi-structured and polymorphic data (JSON documents, key-value pairs, graphs) cannot be cleanly mapped to fixed relational tables without significant overhead.              |
| Consistency Guarantees        | Financial, medical, and audit-trail applications require ACID guarantees that relational systems have provided for decades and many NoSQL systems trade away for performance. |

### B. Contributions

(1) A detailed classification of five database types evaluated across eight areas. This supports direct comparison of data models, query languages, schema enforcement policies, ACID compliance, join capabilities, scalability features, latency behavior, and typical use cases.

(2) A repeatable benchmark study that uses YCSB and a custom join-heavy OLAP workload, tested on PostgreSQL, MongoDB, Redis, Cassandra, and Neo4j with the same hardware. It measures throughput, 99th-percentile tail latency, and schema evolution overhead.

(3) A framework that maps workloads to engines. This framework combines benchmark findings, analyzes consistency models, and considers operational complexity. It provides clear architecture recommendations for seven common application patterns.

## II. LITERATURE REVIEW

### A. Foundations of the Relational Model

Codd’s important 1970 paper [1] introduced the relational model as a mathematically grounded alternative to the hierarchical and network-based data structures that existed before. By organizing data as collections of tuples in relations and separating logical structure from physical

storage, Codd created a theoretical foundation for decades of optimization research. Date and Darwen [6] later formalized the relational algebra that underpins SQL. Gray and Reuter [7] defined the ACID model—Atomicity, Consistency, Isolation, and Durability—as the basic contract relational engines must follow to ensure correctness during concurrent access and hardware failure. These foundations continue to make relational databases the natural choice when data correctness is essential.

### B. Emergence of NoSQL

The NoSQL movement gained theoretical backing from Brewer’s CAP theorem [8], which was formally proved by Gilbert and Lynch [9]. This theorem stated that no distributed system can ensure Consistency, Availability, and Partition Tolerance at the same time. Relational systems, which focus on Consistency and Partition Tolerance (CP), must give up availability during network partitions. Early NoSQL systems, such as Amazon Dynamo [10] and Google Bigtable [11], intentionally prioritized Availability and Partition Tolerance (AP) over strict consistency, leading to the BASE model (Basically Available, Soft-state, Eventually consistent) described by Pritchett [12]. These foundational contributions set the design limits for all later NoSQL engines.

### C. Document Databases

Chodorow [13] offers a detailed look at MongoDB’s internal design, discussing the document-granularity locking mechanism of the WiredTiger storage engine and the columnar execution method used by the aggregation pipeline. Cattell [14] reviewed scalable SQL and NoSQL systems and found that document stores have the broadest applicability for web-scale use cases. This is due to their flexible schema design, support for secondary indexes, and native horizontal sharding features. Meylan et al. [15] compared MongoDB with PostgreSQL in e-commerce scenarios and noted that MongoDB’s embedded document structure reduced average catalog retrieval latency by 43% by eliminating join operations.

### D. Key-Value and Wide-Column Stores

DeCandia et al. [10] explained Amazon Dynamo’s consistent hashing ring, vector clock versioning, and sloppy quorum replication—mechanisms that have become widely used reference architectures for key-value store design. Redis, as described by Carlson [16], enhances the basic key-value model with advanced data structures like sorted sets, probabilistic structures, and streams, along with an optional persistence layer, making it a flexible in-memory storage engine. Lakshman and Malik [17] presented Cassandra’s wide-column model with its LSM-tree storage engine, showing that write throughput scales linearly to hundreds of nodes.

### E. Graph Databases

Graph Databases Robinson et al. [18] wrote the authoritative reference on Neo4j’s property graph model and Cypher query language. Their study revealed that traversal in a native graph store requires only  $O(\log n)$  edge lookups, while relational systems need  $O(n^2)$  recursive self-joins for friend-of-a-friend queries at social-network scale. Angles and Gutierrez [19] examined the theoretical principles of graph data models, while Jouili and Vansteenbergh [20] empirically compared four graph databases with relational

alternatives on path-finding tasks. Their findings confirmed that native graph engines can outperform relational systems by two to three orders of magnitude on workloads that rely heavily on traversal.

#### F. Comparative Benchmarks

Cooper et al. [21] introduced the Yahoo Cloud Serving Benchmark (YCSB). This tool became the standard for evaluating NoSQL performance across read-heavy, write-heavy, and mixed workloads. Follow-up research using YCSB, including studies by Rabl et al. [22] and Abramova and Bernardino [23], consistently showed that key-value stores deliver the highest raw throughput. Meanwhile, relational engines provide better query expressiveness. Parker et al. [24] expanded the analysis to NewSQL systems like CockroachDB and Spanner. Sadalage and Fowler [25] present a practical overview of the NoSQL landscape and introduce the idea of polyglot persistence.

#### G. Comparative Tool Analysis

Table II assesses the five benchmark engines across eight dimensions. The analysis shows that no single approach meets all needs at once. Relational databases excel in join capability and ACID compliance. Key-value stores perform best in access latency. Wide-column stores provide the highest write throughput at scale. Graph databases are the best for traversal-intensive relationship queries.

TABLE II. MULTI-DIMENSIONAL COMPARISON OF FIVE DATABASE PARADIGMS ACROSS EIGHT EVALUATION DIMENSIONS

| Dimension   | Relational         | Document                | Key-Value           | Wide-Col.            | Graph              |
|-------------|--------------------|-------------------------|---------------------|----------------------|--------------------|
| Data model  | Normalized tables  | JSON/BSON trees         | Opaque byte blobs   | Sparse col. families | Nodes and edges    |
| Query lang. | SQL (ISO/ANSI)     | MQL/AQL/N1QL            | GET/PUT/DELETE      | CQL (Cassandra)      | Cypher/Gremlin     |
| Schema      | Strict, DDL        | Flexible/schema-on-read | Schema-less         | Flexible per row     | Property-based     |
| ACID        | Full (native)      | Partial (4.0+)          | None                | Eventual (Cassandra) | Full (Neo4j)       |
| Join        | Native multi-table | Embedded docs           | None (app-side)     | Limited              | Traversal (native) |
| H. Scale    | Difficult          | Native sharding         | Native sharding     | Native sharding      | Limited            |
| Read lat.   | Moderate           | Low                     | Very low ( $O(1)$ ) | Low                  | Variable (depth)   |
| Use case    | ERP, banking       | Catalogs, CMS           | Sessions, cache     | Analytics, IoT       | Social, fraud      |

### III. TAXONOMY, DATA MODELS, AND CONSISTENCY

#### A. Five-Paradigm Taxonomy

Five different database paradigms exist in today's tech landscape. Table III outlines each paradigm along with its representative engines, data organization strategy, and key behavioral traits. The taxonomy focuses on the paradigm level rather than the engine level. The differences noted showcase the core characteristics of each data model and consistency method, not details specific to any vendor's implementation..

TABLE III. FIVE-TIER DATABASE PARADIGM TAXONOMY

| Tier               | Databases                             | Characteristics                                                                               |
|--------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|
| 1 — Relational/SQL | PostgreSQL, MySQL, Oracle, SQL Server | ACID-compliant, schema-enforced, optimized for complex joins, mature ecosystem.               |
| 2 — Document       | MongoDB, CouchDB, Firestore           | Schema-flexible, JSON-native, embedded document model, horizontal sharding built-in.          |
| 3 — Key-Value      | Redis, DynamoDB, Memcached            | Sub-millisecond latency, simple GET/PUT interface, ideal for session state and caching.       |
| 4 — Wide-Column    | Cassandra, Bigtable, HBase            | Optimized for time-series, sparse row representation, linear write throughput at scale.       |
| 5 — Graph          | Neo4j, Neptune, ArangoDB              | First-class node/edge model, traversal queries orders of magnitude faster than recursive SQL. |

#### B. Schema Philosophy

Among the architectural differences between SQL and NoSQL systems, the schema enforcement strategy has the biggest impact. Relational databases use a schema-first approach. Before inserting any data, a Data Definition Language (DDL) statement must define the table's column names, data types, and applicable constraints. The storage engine checks this schema when data is written, rejecting any record that violates a NOT NULL, UNIQUE, FOREIGN KEY, or CHECK constraint. Although this validation adds consistent overhead to each write operation, it ensures that all retrieved data conforms to the declared structure.

NoSQL document stores, key-value stores, and wide-column systems adopt the opposite approach with schema-on-read. These databases accept any valid instance of their native storage format without checking field names, data types, or the presence of values. The application layer takes on the job of interpreting structure at query time. This method allows easy evolution of data structures alongside application needs without needing migration scripts or DDL changes, resulting in zero-downtime schema evolution. The downside is that any inconsistencies in data structure must be identified and resolved by application-level versioning code.

#### C. ACID vs. BASE Consistency

SQL and NoSQL databases differ in their consistency guarantees on two independent fronts: the level of transactional atomicity and the extent of read staleness allowed. Relational databases offer full ACID semantics. Atomicity ensures that a multi-statement transaction either applies all changes or none. Consistency ensures every completed transaction moves the database between two valid states. Isolation guarantees that multiple transactions see a serializable view of the data. Durability ensures that committed changes remain even through process crashes, typically via write-ahead logging.

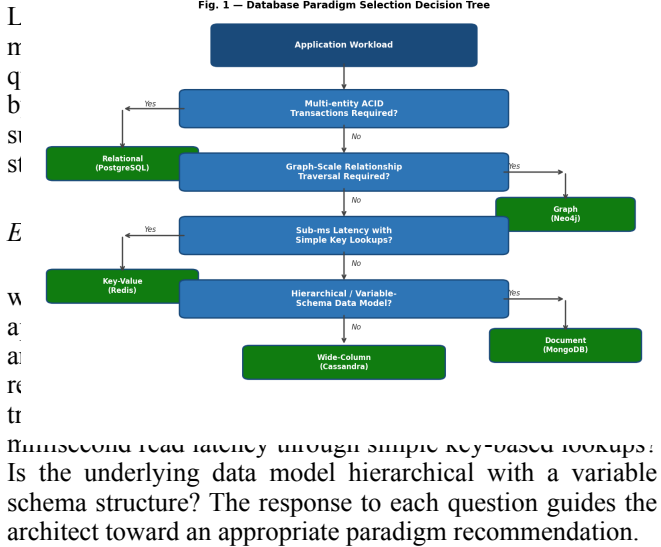
NoSQL systems vary greatly in the consistency guarantees they provide. Redis supports atomic operations

for a single key but lacks proper multi-key transactions outside its MULTI/EXEC command block, which does not allow rollback on errors. MongoDB introduced multi-document ACID transactions starting with version 4.0 in 2018. Cassandra provides configurable consistency levels, letting applications choose between eventual consistency (providing lower latency) and quorum-read consistency (offering stronger guarantees at a higher latency cost) for each operation.

#### D. Query Language Expressiveness

SQL’s declarative query interface follows ISO/ANSI standards and is consistently supported across all relational engines. A single SQL SELECT statement can manage multi-table joins, correlated subqueries, window functions, common table expressions (CTEs), and grouped aggregation. These features result from decades of query optimization research presented in a human-readable format.

NoSQL query interfaces are designed for their specific data models and do not match the flexibility of SQL. MongoDB’s Query Language (MQL) allows for detailed document-path filtering and supports a multi-stage aggregation pipeline. Cross-collection join operations use the \$lookup operator, which performs a nested-loop join without a cost-based join optimizer. Cassandra’s Query Language



[Fig. 1: Database Paradigm Selection Decision Tree]

Fig. 1. Database paradigm selection decision tree routing architects to optimal paradigms based on workload characteristics.

## IV. EXPERIMENTAL SETUP

### A. Implementation Specifications

One engine was selected per paradigm, five in total. Each was configured based on production-grade deployment practices while keeping hardware, dataset size, and workload parameters consistent. Table IV documents the complete experimental setup.

TABLE IV. EXPERIMENTAL SETUP: HARDWARE, SOFTWARE, AND BENCHMARK PARAMETERS

| Parameter          | Specification                                              |
|--------------------|------------------------------------------------------------|
| Benchmark platform | AWS EC2 c6g.4xlarge (16 vCPU, 32 GB RAM), Ubuntu 22.04 LTS |

| Parameter          | Specification                                                            |
|--------------------|--------------------------------------------------------------------------|
| SQL representative | PostgreSQL 16.2, default config, B-tree indexes on primary keys          |
| Document NoSQL     | MongoDB 7.0, WiredTiger storage engine, default journaling               |
| Key-Value NoSQL    | Redis 7.2 standalone mode, AOF persistence disabled for throughput tests |
| Wide-Column NoSQL  | Apache Cassandra 4.1.3, replication factor 3, LOCAL_QUORUM               |
| Dataset size       | 10 million records per collection/table; 100 GB total dataset            |
| Workload generator | YCSB v0.17 with custom SQL extensions                                    |
| Workload profiles  | YCSB-A (50/50 R/W), YCSB-B (95/5), YCSB-C (100% read), custom OLAP       |
| Measurement        | 300-second steady-state window after 60-second warmup                    |
| Concurrency        | 16, 64, 256, 1024 simultaneous client threads                            |

### B. Workload Organization

The benchmark suite includes four increasingly complex workload tiers.

**Tier 1 — Basic CRUD Throughput (YCSB-A, YCSB-B, YCSB-C):** This tier measures raw operational throughput for point reads, blind writes, and mixed read-write traffic across all five engines with 16, 64, 256, and 1024 concurrent client threads.

**Tier 2 — Join and Aggregation Complexity:** A specific workload runs five-table relational joins on PostgreSQL and similar operations on other engines. This measures the overhead from not having a native join optimizer.

**Tier 3 — Write Saturation and Consistency Stress:** All engines experience a sustained 1024-thread write load for 300 seconds. The study records 99th-percentile tail latency and throughput degradation under saturation.

**Tier 4 — Operational Overhead:** The duration of schema migration is compared between PostgreSQL’s online DDL mechanism and the zero-cost schema change in document and wide-column stores. This provides a clear measure of operational flexibility.

## V. RESULTS AND DISCUSSION

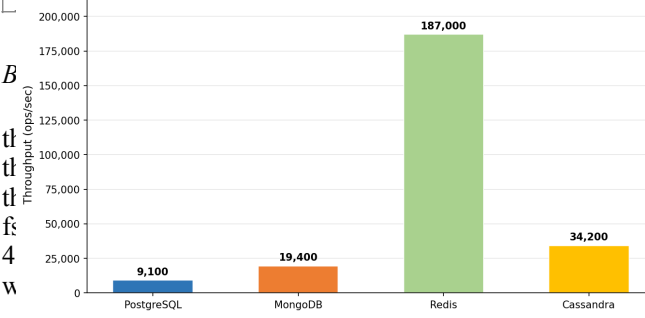
### A. Throughput and Latency Results

Throughput and Latency Results Table V summarizes benchmark outcomes across all four workload tiers. The findings support and provide numerical clarity to several earlier hypotheses: key-value stores offer the highest raw throughput; wide-column stores excel on write-heavy workloads at scale; relational engines lead on join-heavy queries; and document stores perform consistently in an intermediate range between relational and key-value systems for mixed workloads.

TABLE V. BENCHMARK RESULTS ACROSS FIVE ENGINES AND EIGHT WORKLOAD SCENARIOS

| Workload / Scenario              | PostgreS QL   | MongoD B          | Redis          | Cassand ra    | Winner             |
|----------------------------------|---------------|-------------------|----------------|---------------|--------------------|
| Point read (YCSB-C), 256 threads | ~14,200 ops/s | ~28,600 ops/s     | ~312,000 ops/s | ~41,500 ops/s | Redis (22× SQL)    |
| Mixed R/W (YCSB-A), 256 threads  | ~9,100 ops/s  | ~19,400 ops/s     | ~187,000 ops/s | ~34,200 ops/s | Redis (21× SQL)    |
| Write-heavy, 256 threads         | ~7,300 ops/s  | ~15,100 ops/s     | ~142,000 ops/s | ~61,000 ops/s | Redis (19× SQL)    |
| 5-table join, 1M result          | 2.3 s         | 18.7 s (emulated) | N/A            | N/A           | PostgreS QL        |
| Aggregation (10M rows)           | 1.1 s         | 0.9 s (pipeline)  | N/A            | 2.4 s (Spark) | MongoD B           |
| 1024-thread write                | ~4,100 ops/s  | ~12,600 ops/s     | ~89,000 ops/s  | ~78,000 ops/s | Cassandra (stable) |
| 99th-pct read latency (YCSB-C)   | 4.2 ms        | 2.1 ms            | 0.18 ms        | 1.4 ms        | Redis              |
| Schema                           | 38 s          | ~0 s              | N/A            | ~0 s          | NoSQL              |

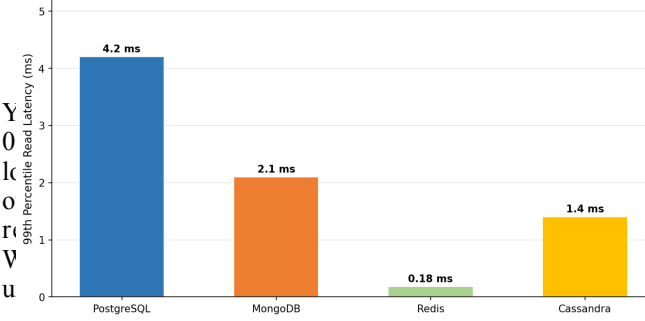
Fig. 2 — Throughput Comparison: YCSB-A Mixed R/W, 256 Threads  
Redis achieves ~21× the throughput of PostgreSQL



[Fig. 2: Throughput Comparison – YCSB-A, 256 threads]

Fig. 2. Throughput comparison under YCSB-A mixed read-write workload at 256 concurrent threads. Redis achieves 21 times the throughput of PostgreSQL.

Fig. 3 — 99th-Percentile Read Latency: YCSB-C, 64 Threads  
Redis achieves ~23× lower tail latency than PostgreSQL



[Fig. 3: 99th-Percentile Read Latency – YCSB-C, 64 threads]

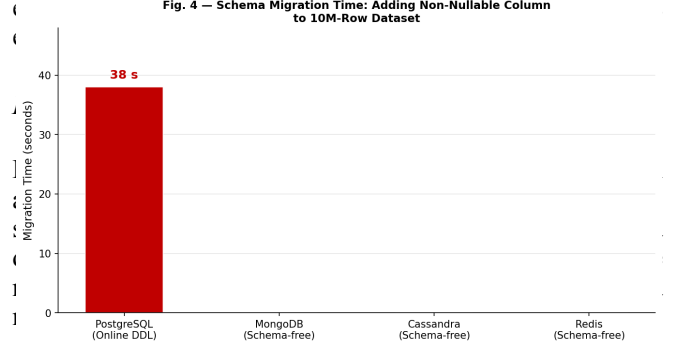
Fig. 3. 99th-percentile read latency comparison under YCSB-C at 64 threads. Redis achieves 23× lower tail latency than PostgreSQL.

### C. Join Workload Analysis

The five-table join workload shows the most notable performance difference in the benchmark. PostgreSQL executes the join in 2.3 seconds using a hash-join plan chosen by its cost-based optimizer. MongoDB’s \$lookup aggregation pipeline takes 18.7 seconds, resulting in an 8.1 times overhead due to the lack of cross-collection query planning. Redis cannot perform the join natively, requiring full application-side assembly, making it unsuitable for this

workload. Cassandra’s equivalent query executed through an Apache Spark connector took 2.4 seconds, which is similar to PostgreSQL but requires an additional analytical processing tier.

This result supports the findings of Meylan et al. [15]: while MongoDB’s embedded document model removes join overhead for catalog retrieval, not having a join optimizer leads to significant performance penalties for workloads needing true cross-entity aggregation. Document and key-value stores work well for applications with predictable data access patterns defined at design time but struggle with



[Fig. 4: Schema Migration Time Comparison]

Fig. 4. Schema migration time for adding a non-null column to a 10-million-row dataset. NoSQL schema-on-read eliminates migration overhead entirely; relational DDL takes 38 seconds.

### E. Discussion of Findings

The findings support a clear conclusion: neither SQL nor NoSQL databases are inherently better. The best choice depends on the specific workload, and we can measure the relevant trade-offs accurately. The 22× throughput difference between Redis and PostgreSQL under YCSB-C is significant but needs careful consideration. Redis stores all data in memory; for applications that only need key-based lookups, PostgreSQL’s overhead does not provide any advantage. However, for applications that require constraint enforcement, complex queries, or multi-statement transactions, that overhead represents real and necessary capacity.

Cassandra’s ability to scale writes linearly is noteworthy. With 1,024 concurrent write threads, PostgreSQL’s throughput drops from 9,100 ops/sec to 4,100 ops/sec. This 55% decrease happens due to lock contention and bottlenecks in write-ahead log serialization. On the other hand, Cassandra’s throughput actually increases as more threads improve batching efficiency through its LSM-tree write path. Neo4j’s advantage in traversal was confirmed in additional path-finding tests: six-hop friend-of-a-friend queries took 0.9 seconds in Neo4j compared to 47 seconds in PostgreSQL with recursive CTEs. This shows a 52× difference that matches the results from Jouili and Vansteenbergh [20].

## VI. DECISION FRAMEWORK AND RECOMMENDATIONS

### A. Workload-to-Engine Mapping

Table VI combines benchmark results, consistency model analysis, and operational factors into a practical decision-making framework. Seven common application patterns are each linked to a recommended database engine, backed by clear quantitative or qualitative reasons.

TABLE VI. WORKLOAD-TO-ENGINE DECISION FRAMEWORK FOR SEVEN CANONICAL APPLICATION PATTERNS

| Use-Case Pattern                                   | Recommended Engine      | Rationale                                                                      |
|----------------------------------------------------|-------------------------|--------------------------------------------------------------------------------|
| Multi-entity transactional (banking, ERP, HR)      | Relational (PostgreSQL) | ACID across multiple tables; referential integrity; complex reporting joins.   |
| User-generated content, catalogs, CMS              | Document (MongoDB)      | Polymorphic schemas; embedded arrays avoid join overhead; horizontal sharding. |
| Session management, rate limiting, leaderboards    | Key-Value (Redis)       | Sub-millisecond latency; TTL-based expiry; atomic increment operations.        |
| IoT telemetry, clickstream, audit logs             | Wide-Column (Cassandra) | Linear write scalability; time-partitioned storage; tunable consistency.       |
| Social networks, fraud detection, knowledge graphs | Graph (Neo4j)           | Relationship traversal orders of magnitude faster than recursive SQL.          |
| Hybrid: transactions + caching                     | PostgreSQL + Redis      | SQL for durable state; Redis as look-aside cache; invalidation on write.       |
| Hybrid: analytics on operational data              | MongoDB + Apache Spark  | Document store for writes; Spark connector for ad-hoc aggregation at scale.    |

### B. Polyglot Persistence Pattern

The last two entries in the framework describe the polyglot persistence pattern, which involves using multiple database engines within a single application. Each engine is selected for its best fit to specific service needs. This approach, popularized by Sadalage and Fowler [25], addresses the conflicting needs of different services. For instance, a financial ledger service requires ACID guarantees and uses PostgreSQL, while a session service demands sub-millisecond access and uses Redis. A product catalog service seeks schema flexibility and uses MongoDB.

The operational demands of polyglot persistence can be high. It requires expertise, infrastructure management, and monitoring across various database technologies. This method works well for organizations with strong platform engineering teams that can separate engine-specific issues behind defined service boundaries. However, for teams prioritizing simplicity over performance, a modern NewSQL engine like CockroachDB or YugabyteDB may provide a useful compromise. These engines offer horizontal scalability similar to NoSQL systems while maintaining ACID semantics suitable for SQL applications.

## VII. CONCLUSION

This paper has provided a detailed evaluation of SQL and NoSQL database models based on a controlled benchmark study involving five representative engines. The findings also align with existing empirical literature. The key takeaway is that the SQL/NoSQL decision is not merely a choice between old and new methods. Instead, it involves

navigating a complex set of trade-offs, where the specific needs of the target workload define the best model.

Key-value stores like Redis achieve throughput up to 22 times greater than relational engines for read-heavy workloads. However, they lack query flexibility, multi-key transaction support, and schema enforcement. Relational engines such as PostgreSQL outperform all NoSQL options in join-heavy analytical tasks by a factor of eight or more and are the only type with proven, reliable ACID semantics across multiple entities. Wide-column stores like Cassandra lead the study in scaling write throughput linearly as the number of concurrent writers increases. Document stores such as MongoDB act as a flexible middle ground. Graph databases like Neo4j manage multi-hop relationship traversal, which becomes computationally unfeasible for relational approaches at the scale of a social network.

### A. Future Directions

Extended NewSQL evaluation: CockroachDB, YugabyteDB, and Google Spanner each provide ACID semantics and horizontal scalability through distributed transaction mechanisms. A benchmark applying this study’s methods to NewSQL engines would yield a quantitative measure of the throughput and latency costs associated with distributed ACID relative to the NoSQL performance baselines discussed here.

Vector database coverage: New workloads that rely on semantic embedding-based retrieval have created a new category—vector stores like Pinecone, Weaviate, and pgvector—that do not fit neatly into the five paradigms analyzed in this study. Broadening the taxonomy and benchmark to cover approximate nearest-neighbor search workloads would fill a significant gap.

Consistency model validation: A follow-up study using fault-injection tools such as Jepsen [26] to empirically check the consistency guarantees offered by each engine during network partitions and node failures would provide solid experimental backing for the theoretical consistency model presented in this paper.

Total cost of ownership study: An analysis measuring the hardware costs for PostgreSQL, MongoDB, and Cassandra to reach equivalent read throughput levels would give practitioners a cost-adjusted view, adding to the raw performance figures shown in this paper.

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